(iii) $(a^x)^y = a^{xy}$ for all $x,\ y \in \mathbb{R}$,
(iv) $(ab)^x = a^x b^x$ for all $x \in \mathbb{R}$,
(v) $a^x = 1$ if and only if $x = 0$.

1. Let us consider $f(x) = a^x, x \in \mathbb{R}$ where $a = 2$.
   Now $f(x) = 2^x,\ x \in \mathbb{R}$. Let us show that f is one-to-one and onto.
   Suppose $f(u) = f(v)$ for some $u, v \in \mathbb{R}$. Then, we have $2^u = 2^v$, which implies that $\frac{2^u}{2^v} = 1$, $\Rightarrow 2^{u-v} = 1$.
   So, $u - v = 0$ and hence $u = v$. Thus f is a one-to-one function.

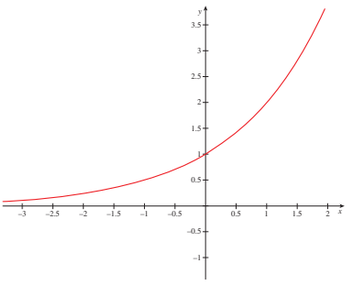


Figure 2.7: $f(x) = 2^x$

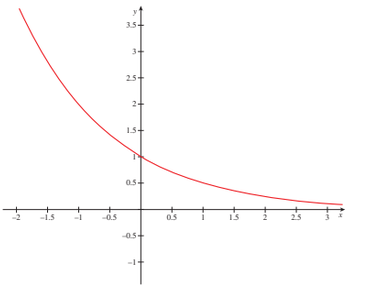


Figure 2.8: $f(x) = \dfrac{1}{2^x}$

   From the graph it is clear that values of $f(x) = 2^x$ increase as x values increase and the range of f is $(0,\ \infty)$. So as $2^0 = 1$, we have $2^x > 1$ for all $x > 0$ and $2^x < 1$ for all $x < 0$. Observe that $f : \mathbb{R} \to (0, \infty)$ is onto.
2. Let us consider $a = \dfrac{1}{2}$. Let $g(x) = \left(\dfrac{1}{2}\right)^x = \dfrac{1}{2^x},\ x \in \mathbb{R}$.
   From the graph it is clear that the values of $g(x) = \left(\dfrac{1}{2}\right)^x$ decrease as x values increase and $g(\mathbb{R}) = (0,\ \infty)$. Also, $g(0) = 1$ we have $g(x) > 1$ for all $x < 0$ and $g(x) < 1$ for all $x > 0$.

Remark: Exactly same arguments as above would show that an exponential function $f(x) = a^x$, for any base $0 < a \neq 1$, is one-to-one and onto with domain $\mathbb{R}$ and codomain $(0,\ \infty)$.

A Special Exponential Function

Among all exponential functions, $f(x) = e^x,\ x \in \mathbb{R}$ is the most important one as it has applications in many areas like mathematics, science and economics. Then what is this e? The following illustration from compounding interest problem leads to the constant e.

Illustration

2.8.3.1 Compound Interest

Recall that if P is the principal, $r = \dfrac{\text{interest rate}}{100}$, n is the number of compounding periods in a year and t is the number of years, then $A = P\left(1 + \frac{r}{n}\right)^{nt}$ gives the total amount after t years. If $n = 4$, then it is compounded quarterly (the interest is added to the existing principal for three months in a year). If $n = 12$, then compounded monthly, $n = 365$ means compounded daily. We can compound every hour, every minute etc. We know that if P and r are fixed and the number of compounding periods in a year increases, then the total amount also increases. Let us consider the case with $P = 1$, $r = 1$ and $t = 1$. Then, we have $A_n = \left(1 + \frac{1}{n}\right)^n$. We want to understand how big it gets as n gets really large. Let us make a table with different values of $n = 10,\ 100,\ 10000,\ 100000,\ 100000000$.